AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X

MODEL EXAMINATION PAPER 2027 AND ONWARDS

Chemistry Paper I

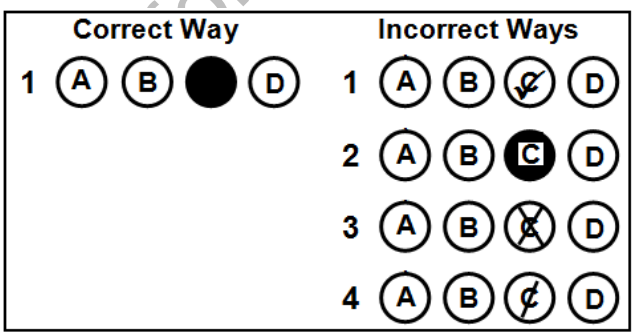
Time: 1 hour 10 minutes Marks: 40

This model paper aims to represent the table of specification with regards to the topics/ content and cognitive demand only.

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INSTRUCTIONS:

1. Check your name and school information on the question paper and Optical Mark Recognition (OMR) sheet. Sign on both if the information is correct.
2. Read each question carefully.
3. Answer all questions on the provided OMR sheet within the allotted time. DO NOT mark your answers on the question paper.
4. There are 100 answer numbers on the OMR sheet. Use answer numbers 1 to 40 only.
5. Each question has four choices: A, B, C, and D. Choose ONE correct answer only. On the OMR sheet, completely fill in the circle corresponding to your choice with a pencil as shown in the example below.



Candidate's Signature

6. If you want to change your answer, ERASE the first answer completely with an eraser, before filling in the new circle. Ensure your marks are dark and clear. Do not make stray marks or leave incomplete fillings as the OMR machine ONLY records what is in the circles.
7. Do not fold, tear, or damage the OMR sheet. Damaged sheets may be scanned incorrectly.
8. You may use a simple calculator if you wish.

1. According to the given chemical equation, the option that can increase the formation of nitrogen dioxide gas is

$N_2O_{4(g)} \leftrightharpoons 2NO_{2(g)}$ $\Delta H > 0$

	Temperature	Pressure
A	high	low
B	low	high
C	high	high
D	low	low

2. The condition that is NOT required for equilibrium is a/ an

A. closed system.
B. constant temperature.
C. equal amounts of reactants and products.
D. equal rates of both forward and backward reactions.

3. The reaction that has the unit of K_c as $mol^{-1}dm^3$ is

A. $SO_{3(g)} + NO_{(g)} \rightleftharpoons SO_{2(g)} + NO_{2(g)}$
B. $2NO_{(g)} + O_{2(g)} \leftrightharpoons 2NO_{2(g)}$
C. $2SO_{3(g)} \rightleftharpoons 2SO_{2(g)} + O_{2(g)}$
D. $N_2O_{4(g)} \leftrightharpoons 2NO_{2(g)}$

4. In a closed container at 900 K, 6.0×10^{-3} M of CH_4 reacts with 4.0×10^{-3} M of H_2O to produce 8.0×10^{-3} M of CO and 3.0×10^{-3} M of H_2. The equation for the reaction is as under.

$CH_{4(g)} + H_2O_{(g)} \rightleftharpoons CO_{(g)} + 3H_{2(g)}$ $K_c = 2.4 \times 10^{-4}$ at 900K

Which of the following combinations is CORRECT as the system proceeds towards equilibrium?

	Relationship of Q_c to K_c	Direction of Net Reaction
A	$Q_c > K_c$	Forward
B	$Q_c < K_c$	Forward
C	$Q_c > K_c$	Backward
D	$Q_c < K_c$	Backward

5. The option that exemplifies an exothermic process is

A. cracking of oil fractions.
B. melting of bituminous coal.
C. burning of petrol in a car engine.
D. fractional distillation of petroleum.

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6. What happens when the following two soluble salts are mixed?

$$BaCl_{2(aq)} + 2AgNO_{3(aq)} \rightarrow ?$$

A. Two soluble salts are produced.
B. Two insoluble salts are produced.
C. One insoluble salt, water and a gas are produced.
D. One soluble salt and one insoluble salt are produced.

7. The given chemical reaction exemplifies

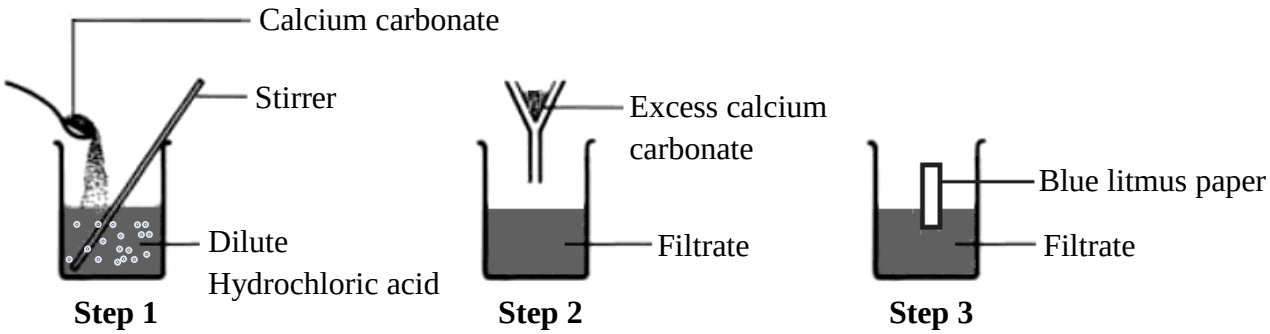
$$Ca(OH)_{2(aq)} \overset{Water}{\rightleftharpoons} Ca^{2+}_{(aq)} + 2OH^{-}_{(aq)}$$

A. Lewis concept.
B. Arrhenius concept.
C. Faraday's concept.
D. Bronsted-Lowry concept.

8. If the hydroxyl ion concentration of a solution is 1×10^{-12}, then the nature of the solution would be

A. highly basic.
B. highly acidic.
C. slightly basic.
D. slightly acidic.

9. A student performs a reaction in three steps as follows and carefully records his observations.



The option that determines the results of step **1** and **3** is

	Step 1	**Step 3**
A	chlorine bubbles off	no change
B	chlorine bubbles off	blue litmus paper turns red
C	carbon dioxide bubbles off	no change
D	carbon dioxide bubbles off	blue litmus paper turns red

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10. Equimolar quantities of potassium chloride and hydrated magnesium chloride are mixed, forming a saturated solution by heating. When the mixture is allowed to stand, carnallite ($KMgCl_3 \cdot 6H_2O$) is produced by slow crystallisation at 25°C.

    Based on the given information, carnallite is a type of a/ an

    A. acidic salt.
    B. double salt.
    C. normal salt.
    D. complex salt.

11. When ethyne is bubbled through ammoniacal silver nitrate solution, a yellow-white precipitate of silver acetylide is formed.

    The formation of precipitate indicates that in a molecule of ethyne the

    A. terminal hydrogen atom is basic in nature.
    B. terminal hydrogen atom is acidic in nature.
    C. electrons in the triple bond are easily available.
    D. electrons in the double bond are easily available.

12. Under normal conditions, alkanes do not react with strong oxidising agents such as potassium permanganate ($KMnO_4$). This is because of all of the following reasons EXCEPT that alkanes

    A. have non-polar bonds.
    B. give substitution reactions.
    C. are saturated hydrocarbons.
    D. are soluble in polar solvents.

13. C_2H_5OH can be converted into C_2H_4 by the process of

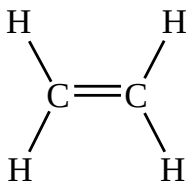
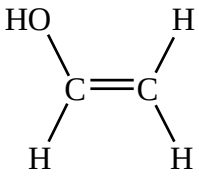
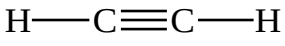
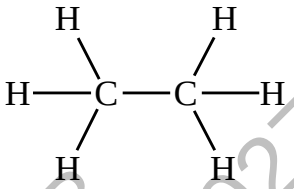
    A. reduction.
    B. dehydration.
    C. neutralisation.
    D. polymerisation.

14. The complete combustion of methane in the presence of two moles of oxygen gas releases

    A. 2 moles of water, 1 mole of carbon and heat.
    B. 2 moles of hydrogen gas, 1 mole of carbon and heat.
    C. 2 moles of water, 1 mole of carbon dioxide and heat.
    D. 2 moles of hydrogen gas, 1 mole of carbon dioxide and heat.

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15. A substitution reaction takes place when bromine reacts with

	
A	B
	
C	D

16. Alkanes, alkenes and alkynes are classified as organic compounds. This is because they all

A. are hydrocarbons.
B. are saturated molecules.
C. have high melting points.
D. have high rate of reaction.

17. $H_2C = CH - CH = CH - CH_3$

The IUPAC (International Union of Pure and Applied Chemistry) name of the given organic compound is

A. 1,3-pentene.
B. 2,4-pentene.
C. 1,3-pentadiene.
D. 2,4-pentadiene.

18. A compound that contains ester functional group is

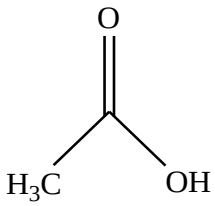
A. CH_3CH_2CHO
B. CH_3CH_2COOH
C. $CH_3COCH_2CH_3$
D. $CH_3COOCH_2CH_3$

19. If CH_3COCH_3 is compared with $CH_3OCH_2CH_3$, it would be concluded that they both

A. have same molecular formula.
B. are chain isomers of each other.
C. have different functional groups.
D. belong to the same homologous series.

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20. The functional group present in the given molecule of vinegar is the



A. ketone.
B. alcohol.
C. aldehyde.
D. carboxylic acid.

21. Compared to ribonucleic acid (RNA), a molecule of deoxyribonucleic acid (DNA) is

I. single stranded
II. self-replicating
III. located in mitochondria

A. I only.
B. I and II.
C. III only.
D. II and III.

22. Maltose is a disaccharide in which two molecules of glucose join by a/ an

A. ester linkage.
B. peptide bond.
C. glycosidic linkage.
D. phosphodiester bond.

23. A set of reactions that is involved in the formation and breakdown of peptide linkage in proteins is

	Formation of Peptide Linkage	**Breakdown of Peptide Linkage**
A	addition	hydrolysis
B	addition	oxidation
C	condensation	hydrolysis
D	condensation	oxidation

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24. The hydrogenation of vegetable oils in the presence of nickel as a catalyst at 60°C results in the formation of a compound **X** having **Y** bonds.

The option that represents **X** and **Y** is

	X	**Y**
A	saturated fats	single bonds
B	unsaturated fats	single bonds
C	saturated fats	double bonds
D	unsaturated fats	double bonds

25. Amylase enzyme is used in making bread as they facilitate the breakdown of

A. starch.
B. lactose.
C. maltose.
D. cellulose.

26. Permanent hardness in water is due to the presence of chlorides and sulphates of

A. sodium.
B. potassium.
C. aluminium.
D. magnesium.

27. *Naegleria fowleri*, a brain eating microorganism, may survive in swimming pools if these are not properly cleaned, cooled and

A. filtered.
B. flocculated.
C. chlorinated.
D. sedimented.

28. Jaundice is caused due to an increase in the amount of

A. iron in the blood.
B. red cells in the blood.
C. uric acid in the blood.
D. bile pigments in the blood.

29. Which of the following compounds of calcium precipitates out when hard water is treated with slaked lime?

A. Sulphates
B. Chlorides
C. Carbonates
D. Bicarbonates

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30. When excessive nutrients from agricultural run-off add into ponds, the concentration of a gas that decreases in water is

A. oxygen.
B. nitrogen.
C. methane.
D. carbon dioxide.

31. It is advised to switch off gas heaters before going to sleep.

This is because the combustion of natural gas in a poorly ventilated room produces

A. carbon dioxide.
B. sulphur dioxide.
C. nitrogen dioxide.
D. carbon monoxide.

32. Presence of pollutants in air has an adverse effect on the environment.

In order to control air pollution, a country's government should ensure that its citizens AVOID the

A. use of private vehicles.
B. plantation of more trees.
C. use of non-combustible sources of energy.
D. setting up of industries away from residential areas.

33. When decomposers feed on dead organic matter, they return carbon to air as carbon dioxide.

This information shows an interaction between

A. lithosphere and biosphere.
B. biosphere and atmosphere.
C. atmosphere and lithosphere.
D. hydrosphere and lithosphere.

34. An adverse effect of acid rain is

A. greenhouse effect.
B. melting of glaciers.
C. depletion of ozone.
D. death of aquatic life.

35. In the soil, acid rain causes an increase in

I. pH value
II. toxic metals
III. the number of microorganisms

A. I only.
B. II only.
C. I and III.
D. II and III.

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36. In the Haber process, nitrogen and hydrogen are combined to form ammonia.

The set of information that stands TRUE for this process is

	Combination Ratio of Nitrogen : Hydrogen	**Overall Energy Change**
A	1 : 3	Exothermic
B	2 : 3	Endothermic
C	3 : 1	Endothermic
D	3 : 2	Exothermic

37. $CO_{2(g)} + 2NH_{3(g)} \xrightarrow{\text{heat/ pressure}} ?$

The product(s) of the given reaction is/ are

A. $NH_4HCO_{3(s)}$
B. $(NH_4)_2CO_{3(s)}$
C. $NH_2COOH_{(s)} + H_{2(g)}$
D. $(NH_2)_2CO_{(s)} + H_2O_{(g)}$

38. During the extraction of copper, impurities float in the form of slag over the molten mixture of copper compounds.

The slag obtained in this extraction is

A. FeS
B. FeO
C. $FeSiO_3$
D. $CuFeS_2$

39. The set of chemicals used in the recovery of ammonia in Solvay's process is

A. sodium carbonate + ammonium chloride
B. ammonium chloride + calcium carbonate
C. ammonium chloride + calcium hydroxide
D. sodium chloride + ammonium bicarbonate

40. During the extraction of copper, the process of roasting involves the conversion of

A. oxides into silicates.
B. oxides into sulphates.
C. sulphides into oxides.
D. carbonates into oxides.

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